

# Memory-Driven Robotic Arm with WiFi Enabled Control

Vineet Mankani, Chirag Wadhwani, Amol Deshpande

Department of Electronics and Telecommunications Engineering

Sardar Patel Institute of Technology, Mumbai, India

vineet.mankani@spit.ac.in, chirag.wadhwani@spit.ac.in, amol\_deshpande@spit.ac.in

**Abstract**—This project emerges against the backdrop of industries grappling with the challenges posed by manual labor-intensive tasks. Traditionally, industrial processes involving 'pick-up and place' operations have been labor-intensive, prone to human error, and subject to repetitive strain injuries. The demand for increased efficiency, precision, and workplace safety has driven the exploration of advanced robotic solutions to alleviate the burdens associated with routine and monotonous tasks. This Memory-Driven Robotic Arm with WiFi Enabled Control represents a pioneering step toward mitigating the drawbacks of manual labor in industries, offering a transformative solution that combines cutting-edge technology with ease of operation.

**Index Terms**—Robotic Arm, WiFi Control, ESP32, Memory-Driven Technology, Industrial Automation

## I. INTRODUCTION

In the dynamic landscape of modern industry, the pursuit of enhanced productivity and operational efficiency has led to a critical reevaluation of traditional manual labor practices. Industries grappling with 'pick-up and place' tasks face challenges such as human error, physical strain, and the inherent limitations of repetitive processes. Recognizing these challenges, this project presents a pioneering venture into the realm of robotics with the development of a Memory-Driven Robotic Arm featuring WiFi Enabled Control.

Focused on precision, adaptability, and seamless integration, this robotic arm represents a strategic response to the evolving needs of industries seeking to optimize their workflows. Through a fusion of TowerPro SG90 Servo Motors and the versatile ESP32 development board, this project endeavors to usher in a new era where advanced robotics not only alleviates the burden of manual labor but also augments industrial processes with intelligence and precision. This introduction sets the stage for a detailed exploration of the project's objectives, methodology, and outcomes, as it seeks to address the pressing challenges of contemporary industrial practices.

## II. MOTIVATION

Industries face the need for versatile solutions that can adapt to different tasks and environments. The Memory-Driven Robotic Arm, with its configurable design, caters to a range of applications, providing adaptability and versatility to address diverse industrial needs. Industrial processes need to keep pace with technological advancements to remain competitive. By

incorporating WiFi-enabled control and memory-driven capabilities, the project aligns with Industry 4.0 trends, ensuring the robotic arm is advanced and future-proofed for evolving technological landscapes.

## III. PROBLEM FORMULATION

The problem at hand centers on the challenges inherent in labor-intensive tasks within industrial settings, particularly those involving the manual execution of 'pick-up and place' operations. This manual approach not only results in increased operational costs and inefficiencies but also introduces inconsistencies, errors, and safety concerns. The envisioned Memory-Driven Robotic Arm project aims to address these issues by devising a versatile solution capable of autonomously executing tasks with precision while mitigating safety risks associated with manual labor. By confronting the limitations of current practices, this project aspires to introduce a transformative paradigm in industrial workflows, leveraging advanced robotics to enhance efficiency, adaptability, and workplace safety.

Developing an effective solution involves the intricate integration of hardware and software components, namely the Robotic Arm, programming of the motors, and the establishment of a user-friendly interface via smartphone for WiFi control.

## IV. LITERATURE REVIEW

Existing research on robotic arms highlights various design approaches, control mechanisms, and applications, offering valuable insights into the field. In one study [1], a 5-degree-of-freedom robotic arm controlled by an Arduino microcontroller and potentiometers successfully executes pick-and-place operations. Building upon this foundation, the Memory-Driven Robotic Arm with WiFi-Enabled Control introduces pioneering features such as WiFi connectivity, real-time mobile device control, and the ability to record/playback sequences, enhancing versatility and functionality beyond traditional models.

Another notable contribution is an Arduino-controlled robotic arm featuring a soft catching gripper [2] for safer object handling in industrial settings. While safety and efficiency are shared concerns, the Memory-Driven Robotic Arm takes a step further by introducing WiFi-enabled control, enabling dynamic mobile device interactions, and offering



Fig. 1. The CO2 Laser Cutter

recording/playback capabilities. Additionally, the present study emphasizes versatility, allowing the arm to perform diverse pick-and-place tasks, overcoming limitations seen in systems designed for smaller weights. Overall, the innovative features of the Memory-Driven Robotic Arm extend beyond the existing literature, contributing to the advancement of robotic arm capabilities for adaptable and automated applications across various industries.

## V. THE ROBOTIC ARM

The project involves the meticulous design and assembly of the robotic arm, constructed from laser-cut 3 mm acrylic sheet components. The use of Fusion360 software ensures precision, while nuts and bolts facilitate a sturdy assembly of the components. The strategic placement of TowerPro SG90 Servo Motors in the base, shoulder, elbow, and gripper ensures optimal functionality.

### A. Design

The robotic arm's design is a crucial aspect of its functionality, ensuring optimal performance and versatility. The arm is composed of four TowerPro SG90 Servo Motors, strategically placed at the base, shoulder, elbow, and gripper, each with a designated range of motion from 0 to 180 degrees. The design (Fig 2) is meticulously crafted using Fusion360 software, facilitating precise modeling of each part. The fundamental design is obtained from the GRABCAD Community, and tweaked as per our own requirements. A 3 mm thickness acrylic sheet serves as the primary material for the arm components, cut seamlessly using a CO2 Laser Cutter (Fig 1).

The choice of acrylic as the primary material was driven by its ideal balance of durability, flexibility, and ease of fabrication. It allowed for precise laser cutting, ensuring intricate components, while its structural integrity provided a sturdy framework for the robotic arm's operations. The design process emphasizes efficiency, balance, and durability to enable seamless integration with the selected servo motors.

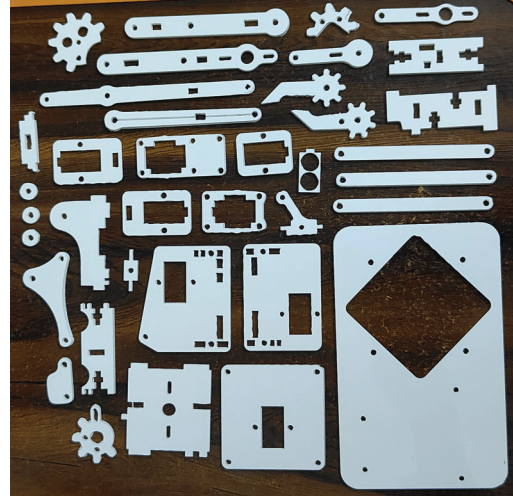


Fig. 2. Robotic Arm 3D Acrylic Parts

The intricate detailing of the components ensures a smooth range of motion, crucial for the arm's intended pick-and-place operations. The resulting design sets the foundation for the subsequent assembly phase.

### B. Assembly

The assembly of the robotic arm involves the integration of designed components into a fully functional and stable structure. The laser-cut acrylic parts are carefully assembled using nuts and bolts, aligning with the precision achieved during the design phase. Each TowerPro SG90 Servo Motor finds its place at the designated locations – the base, shoulder, elbow, and gripper – ensuring a harmonious connection between the mechanical and electronic elements.

The assembly process demands attention to detail to guarantee the proper alignment and secure attachment of the motors. Nuts and bolts provide structural integrity, allowing the arm to withstand the dynamic movements required for its intended tasks. Once assembled, the robotic arm stands as a testament to the successful fusion of design and engineering, poised for the next phase of implementation – programming the motors (Fig 3).

## VI. PROGRAMMING

### A. Driving the servo motors

The programming orchestrates the precise movements of the robotic arm's four TowerPro SG90 Servo Motors using an ESP32 development board. The code initializes and calibrates each servo motor, ensuring accurate and repeatable motions. A WebSocket interface facilitates real-time communication with a mobile device, allowing users to control individual motors—Base, Shoulder, Elbow, and Gripper—through commands (Fig 4). A distinctive feature includes the ability to record and playback sequences of motor movements. Commands such as "Record" initiate the recording of motor positions and "Play" triggers the automated replay of recorded

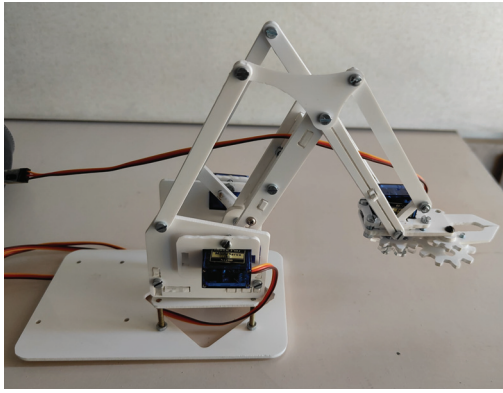


Fig. 3. Assembled Robotic Arm

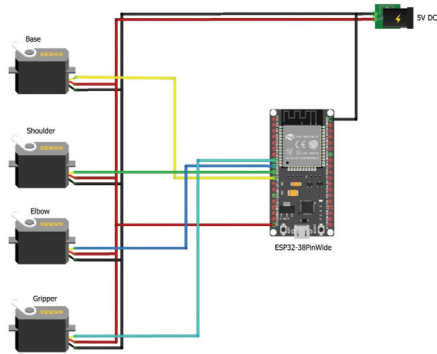


Fig. 4. Connections of ESP32 to the four motors

actions. The code intelligently captures and plays back these sequences, transforming the robotic arm into a dynamic and controllable tool for pick-and-place operations.

This programming section represents the core intelligence of the robotic arm, translating user commands into precise motor actions and enabling automation through recording and playback functionalities.

#### B. ESP32 Development Board

The programming is executed on an ESP32 development board, specifically the NodeMCU-32S, featuring a 30-pin ESP32-WROOM-DA microcontroller (Fig 5). The code is designed to interpret commands from external devices, translating them into precise motor movements. The 4 joints (4 motors) are connected to digital pins 26, 25, 33 and 32 on the ESP32 development board.

The ESP32 microcontroller generates a WiFi soft access point, allowing seamless connectivity with external devices. The user can connect to the robotic arm through a local host URL on a device like the smart phone. This establishes a platform for intuitive control over the arm's movements.

### VII. SMART PHONE CONTROL (WiFi)

The smartphone control aspect of the robotic arm leverages WiFi connectivity to enable seamless interaction between the user's mobile device and the ESP32 development board.

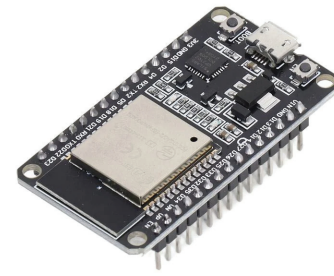


Fig. 5. 30 pin ESP32 Development Board

## Robot Arm Control (using ESP32)

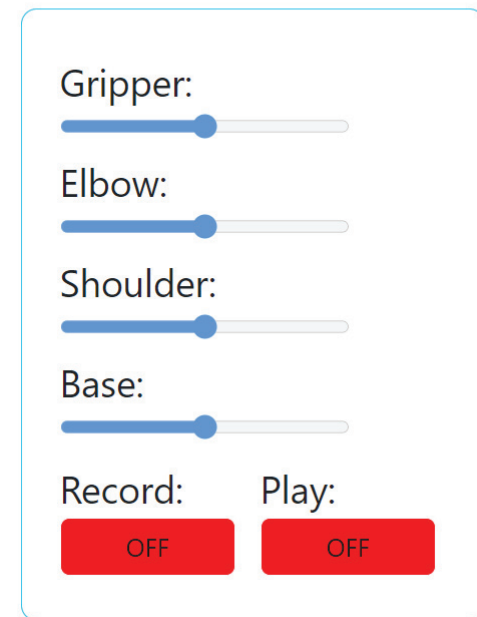


Fig. 6. Driving the motors using the web application through WiFi

Through a user-friendly mobile interface, users can control the robotic arm's movements in real-time (Fig 6).

The ESP32 development board acts as a WiFi access point (AP), allowing users to connect their smartphones to the robotic arm. The access point is set up with the credentials "RobotArm" and a password providing a dedicated connection for control.

The WebSocket communication protocol ensured rapid and responsive interactions between the user's mobile device and the robotic arm. Commands, including motor control, recording, and playback requests, were transmitted seamlessly, highlighting the reliability of the communication system.

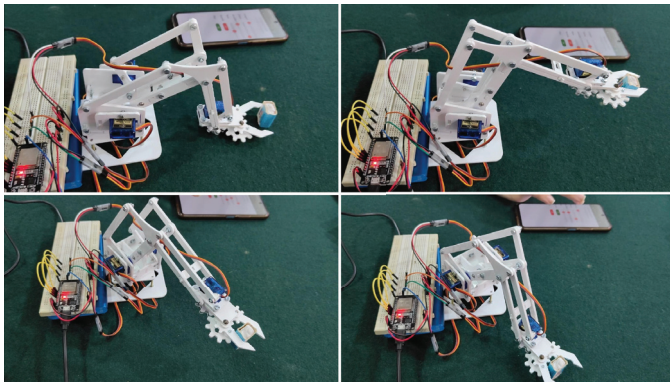


Fig. 7. Robotic Arm picking up and placing an object

## VIII. RESULT

The Memory-Driven Robotic Arm with WiFi-Enabled Control successfully achieved its objectives (Fig 7). The combination of robust mechanical design, precise motor control, and intuitive user interface through WiFi connectivity positions the robotic arm as a versatile and efficient tool for various applications, including pick-and-place tasks in industrial settings. The recorded playback functionality further enhances its utility for repetitive operations, showcasing the potential for automation in practical scenarios. The next steps involve exploring scalability, optimization, and potential applications in diverse industries.

## IX. CONCLUSION

The Memory-Driven Robotic Arm with WiFi-Enabled Control presents a promising prototype for applications in diverse industries. The successful integration of precise motor control, intuitive mobile interface, and recording/playback functionalities positions the robotic arm as a versatile tool for pick-and-place tasks. Its potential applications range from assembly line automation in manufacturing to warehouse logistics, showcasing the adaptability and efficiency of the prototype. The innovative features of real-time control and automated task repetition make it a valuable asset for industries seeking flexible and efficient robotic solutions. Further exploration of scalability and optimization will contribute to its broader adoption in practical settings, marking a significant step towards enhancing automation in various industries.

## ACKNOWLEDGMENT

The authors would like to acknowledge the Nidhi-Prayas lab at Sardar Patel Technology Business Incubator (SP-TBI), for the unwavering support in the design of the robotic arm, which lead to the successful completion of the project.

## REFERENCES

- [1] A. Bhargava and A. Kumar, "Arduino controlled robotic arm," *2017 International conference of Electronics, Communication and Aerospace Technology (ICECA)*, Coimbatore, India, 2017, pp. 376-380, doi: 10.1109/ICECA.2017.8212837.
- [2] Ashly Baby, Chinnu Augustine, Chinnu Thampi, Maria George, Abhilash A P, Philip C Jose, "Pick-and-Place-Robotic-Arm-Implementation-Using-Arduino," *IOSR Journal of Electrical and Electronics Engineering (IOSR-JEEE)*, e-ISSN: 2278-1676,p-ISSN: 2320-3331, Volume 12, Issue 2 Ver. III (Mar. – Apr. 2017), doi: 10.9790/1676-1202033841.
- [3] K. Kruthika, B. M. Kiran Kumar and S. Lakshminarayanan, "Design and development of a robotic arm," *2016 International Conference on Circuits, Controls, Communications and Computing (I4C)*, Bangalore, India, 2016, pp. 1-4, doi: 10.1109/CIMCA.2016.8053274.
- [4] Shyam.R.Nair, " Design of a Robotic Arm for Picking and Placing an Object Controlled Using LABView ", *International Journal for Scientific and Research Publications*, Vol 2, Issue 5, May 2012